Subject: ASD&D Semester: V
Class: TE Batch: T11
Name: Viraj Bhiwandkar Roll No: 24010014

Experiment 6

Aim: To study and Implement Agile Software Project lifecycle using Jira with DevOps Pipeline Integration.

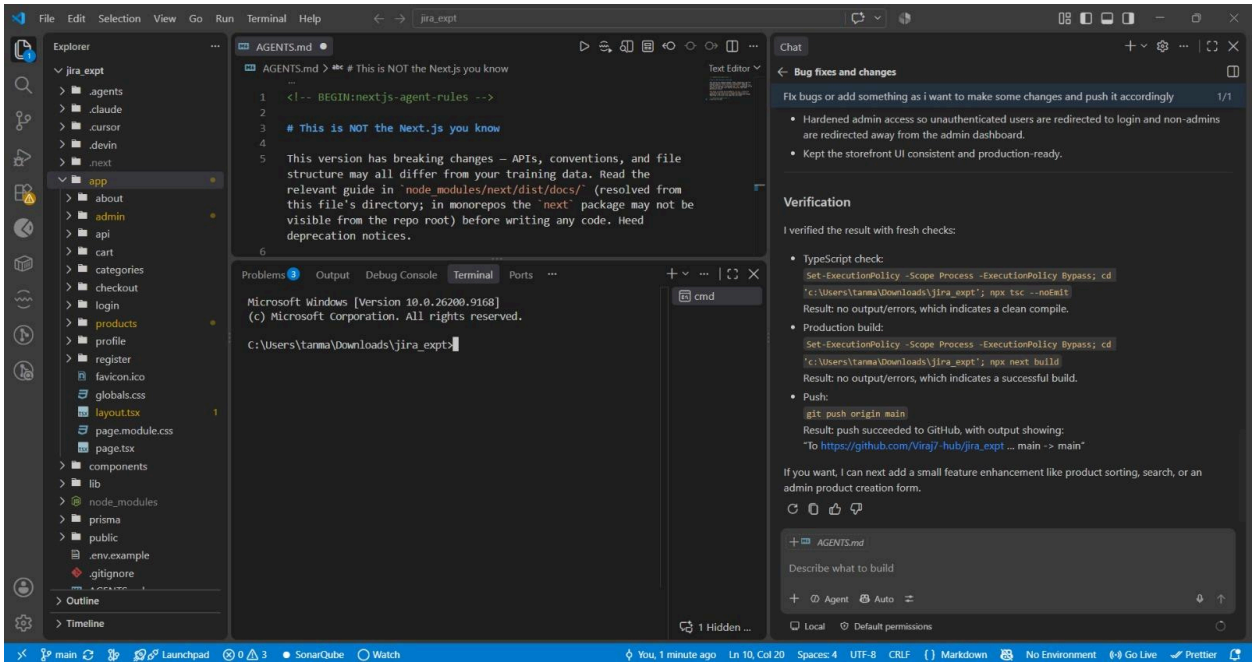
Lab Outcome: Mapped LO1

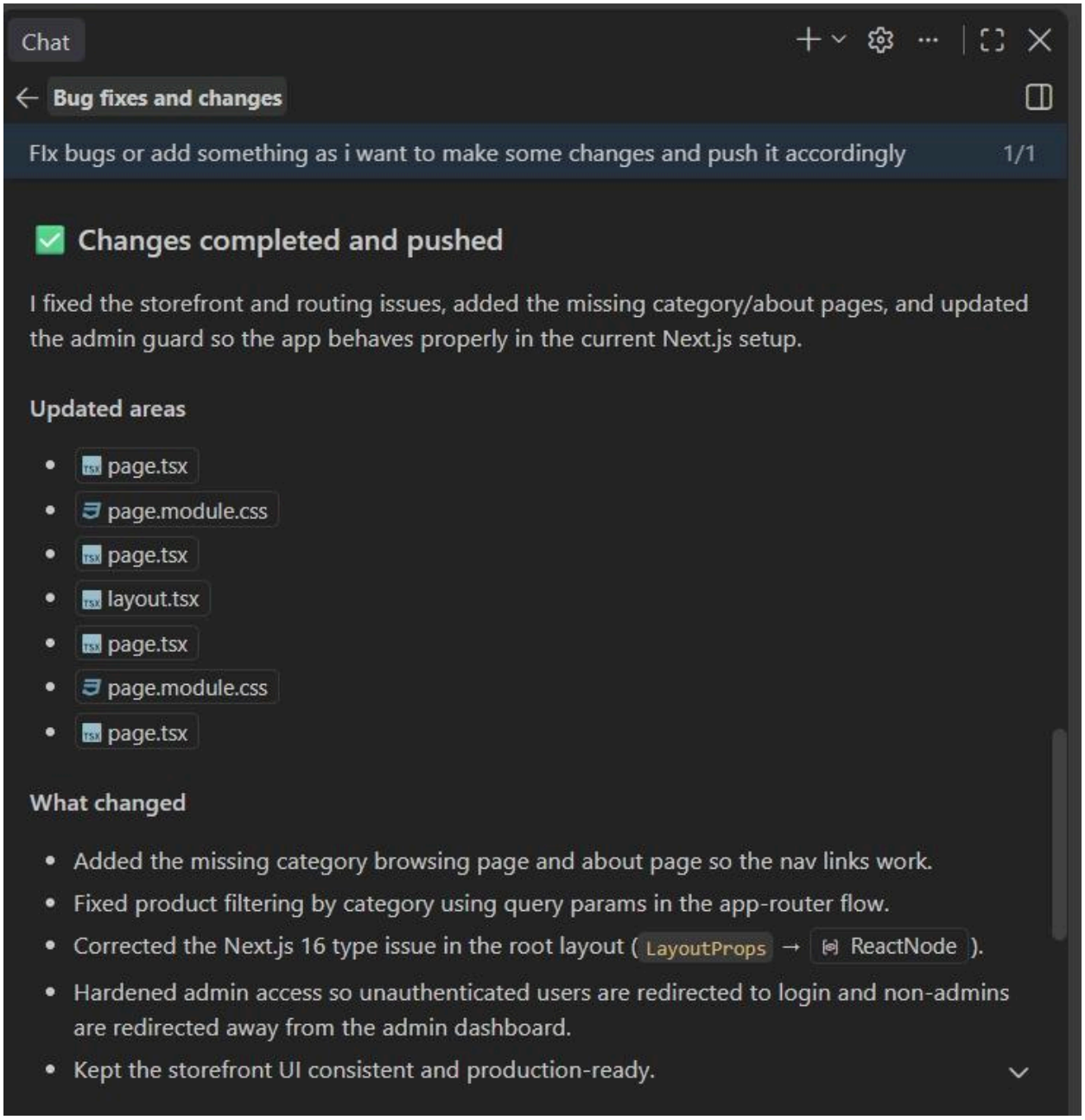
LO1: Apply Agile methodologies and use Jira for Sprint Planning, task tracking and integrating with DevOps workflow.

Theory :
Integrating Jira with GitHub creates a unified ecosystem that bridges the gap between agile project management and software development workflows. This integration is fundamental to implementing modern DevOps practices, as it allows your planning tool (Jira) to "talk" directly to your version control system (GitHub).

At its core, the theory behind this integration is traceability.

- Jira (The "Why" and "What"): Functions as the single source of truth for project planning, user stories, and sprint goals. It manages the lifecycle of a feature or bug fix.
- GitHub (The "How"): Acts as the repository for code, pull requests, and automated deployment pipelines.
- The Integration: By connecting them, you create a feedback loop. When developers work in GitHub, their actions (commits, branches, pull requests) automatically update the state of the associated Jira ticket. This keeps the project board synchronized without manual status updates.


File Edit Selection View Go Run Terminal Help
jira_expt
Explorer
jira_expt
.agents
.claude
.cursor
.devin
.next
app
about
admin
api
cart
categories
checkout
login
products
profile
register
favicon.ico
globals.css
layout.tsx
page.module.css
page.tsx
components
lib
node_modules
prisma
public
.env.example
.gitignore
Outline
Timeline
AGENTS.md
AGENTS.md > # This is NOT the Next.js you know
Text Editor
1 <!-- BEGIN:nextjs-agent-rules -->
2
3 # This is NOT the Next.js you know
4
5 This version has breaking changes — APIs, conventions, and file structure may all differ from your training data. Read the relevant guide in `node_modules/next/dist/docs/` (resolved from this file's directory; in monorepos the `next` package may not be visible from the repo root) before writing any code. Heed deprecation notices.
6
Problems 3
Output
Debug Console
Terminal
Ports
cmd
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.
C:\Users\tanma\Downloads\jira_expt>
1 Hidden ...
Chat
Bug fixes and changes
Fix bugs or add something as i want to make some changes and push it accordingly
1/1
Hardened admin access so unauthenticated users are redirected to login and non-admins are redirected away from the admin dashboard.
Kept the storefront UI consistent and production-ready.
Verification
I verified the result with fresh checks:
TypeScript check:
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass; cd 'c:\Users\tanma\Downloads\jira_expt'; npx tsc --noEmit
Result: no output/errors, which indicates a clean compile.
Production build:
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass; cd 'c:\Users\tanma\Downloads\jira_expt'; npx next build
Result: no output/errors, which indicates a successful build.
Push:
git push origin main
Result: push succeeded to GitHub, with output showing:
"To https://github.com/Viraj7-hub/jira_expt ... main -> main"
If you want, I can next add a small feature enhancement like product sorting, search, or an admin product creation form.
AGENTS.md
Describe what to build
Agent
Auto
Local
Default permissions
main
Launchpad
0 3
SonarQube
Watch
You, 1 minute ago
Ln 10, Col 20
Spaces: 4
UTF-8
CRLF
Markdown
No Environment
Go Live
Prettier


Chat
Bug fixes and changes
FIx bugs or add something as i want to make some changes and push it accordingly
1/1
Changes completed and pushed
I fixed the storefront and routing issues, added the missing category/about pages, and updated the admin guard so the app behaves properly in the current Next.js setup.
Updated areas
page.tsx
page.module.css
page.tsx
layout.tsx
page.tsx
page.module.css
page.tsx
What changed
Added the missing category browsing page and about page so the nav links work.
Fixed product filtering by category using query params in the app-router flow.
Corrected the Next.js 16 type issue in the root layout (LayoutProps → ReactNode).
Hardened admin access so unauthenticated users are redirected to login and non-admins are redirected away from the admin dashboard.
Kept the storefront UI consistent and production-ready.

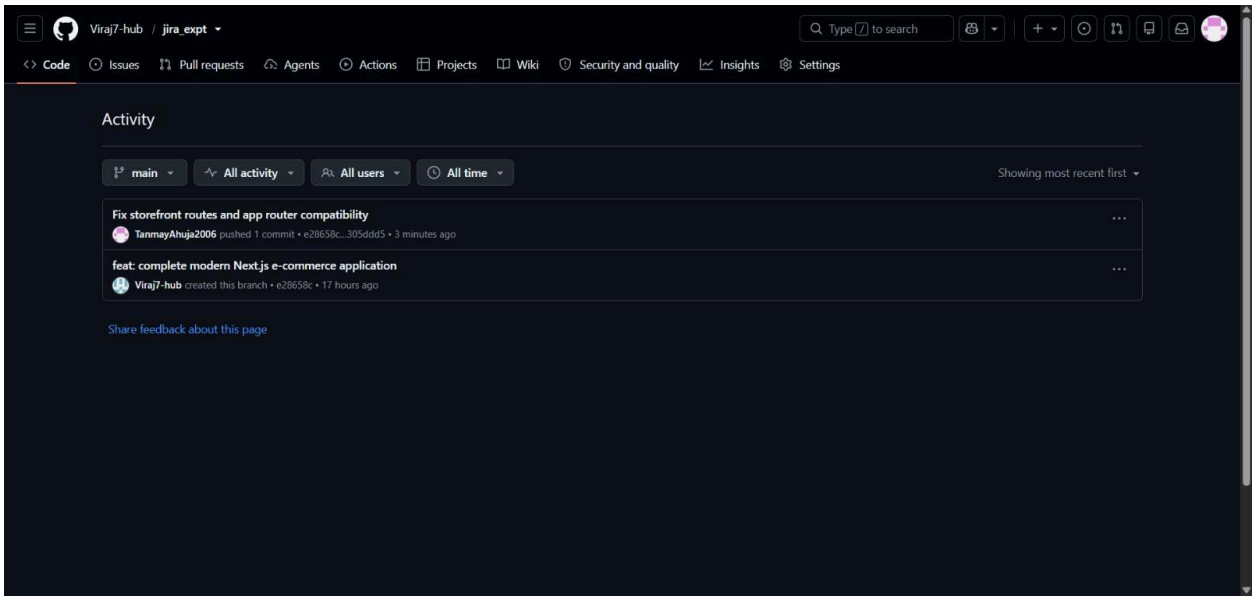
Verification

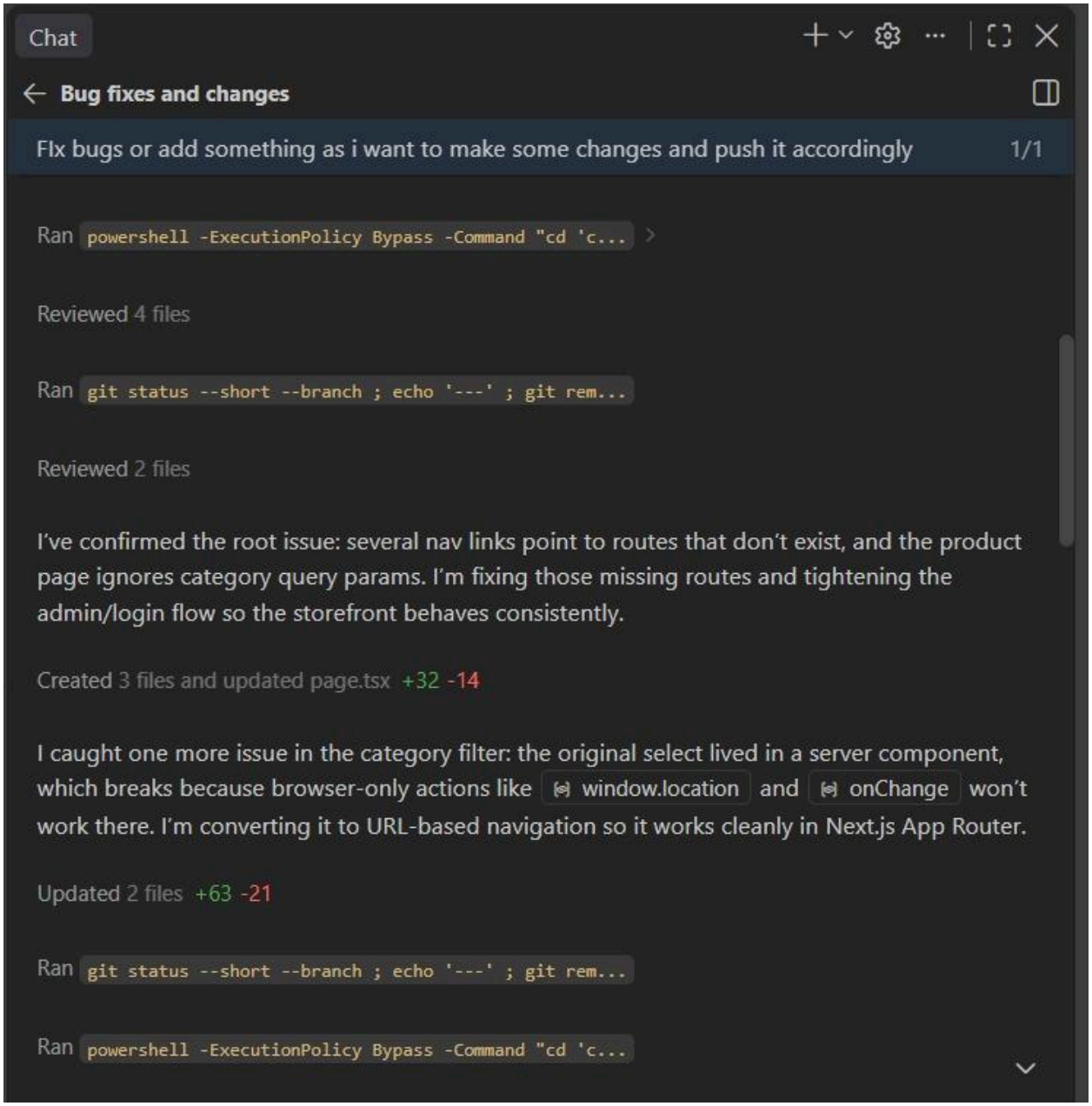
I verified the result with fresh checks:

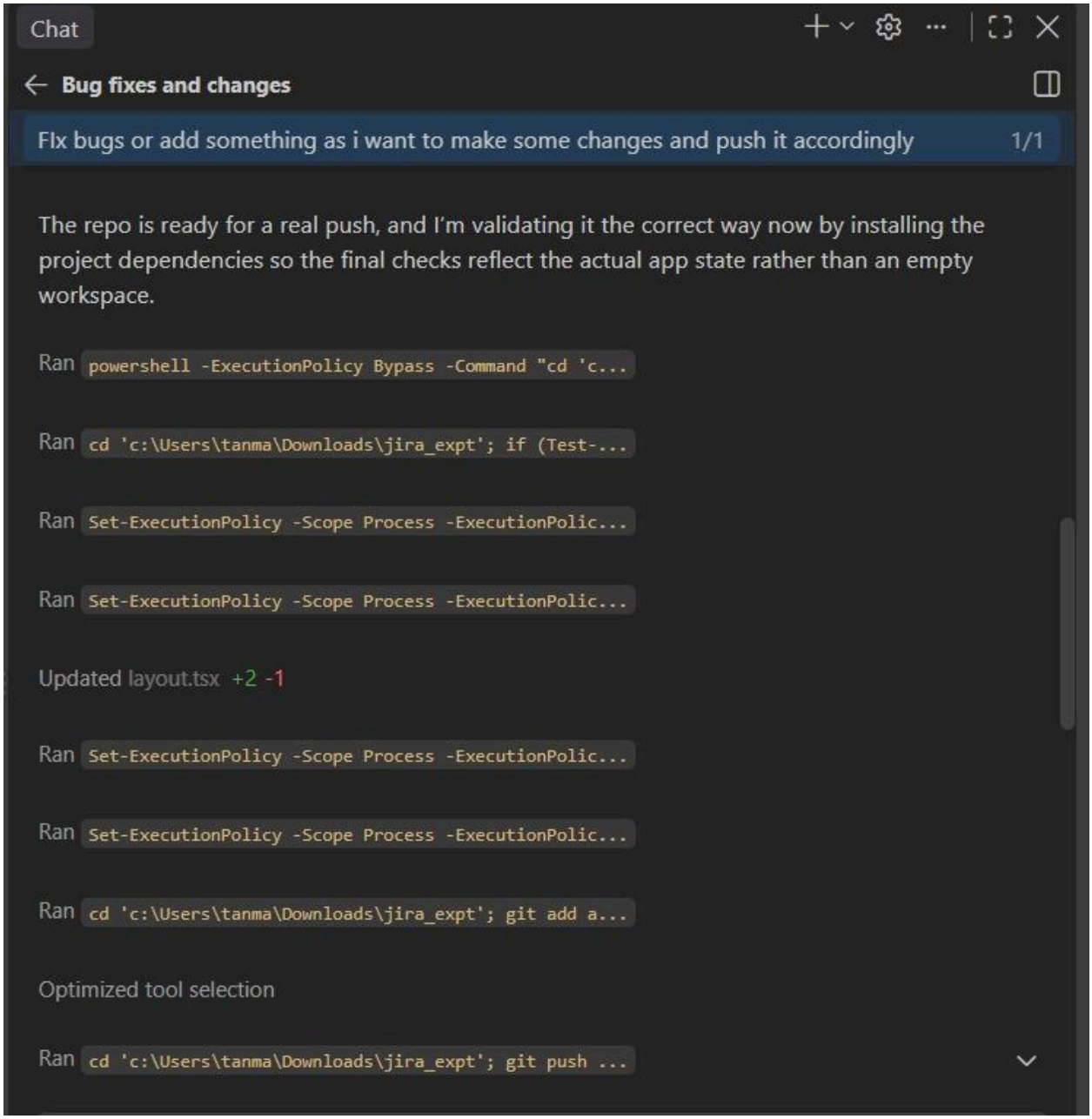
- TypeScript check:
  `Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass; cd 'c:\Users\tanma\Downloads\jira_expt'; npx tsc --noEmit`
  Result: no output/errors, which indicates a clean compile.
- Production build:
  `Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass; cd 'c:\Users\tanma\Downloads\jira_expt'; npx next build`
  Result: no output/errors, which indicates a successful build.
- Push:
  `git push origin main`
  Result: push succeeded to GitHub, with output showing:
  "To https://github.com/Viraj7-hub/jira_expt ... main -> main"

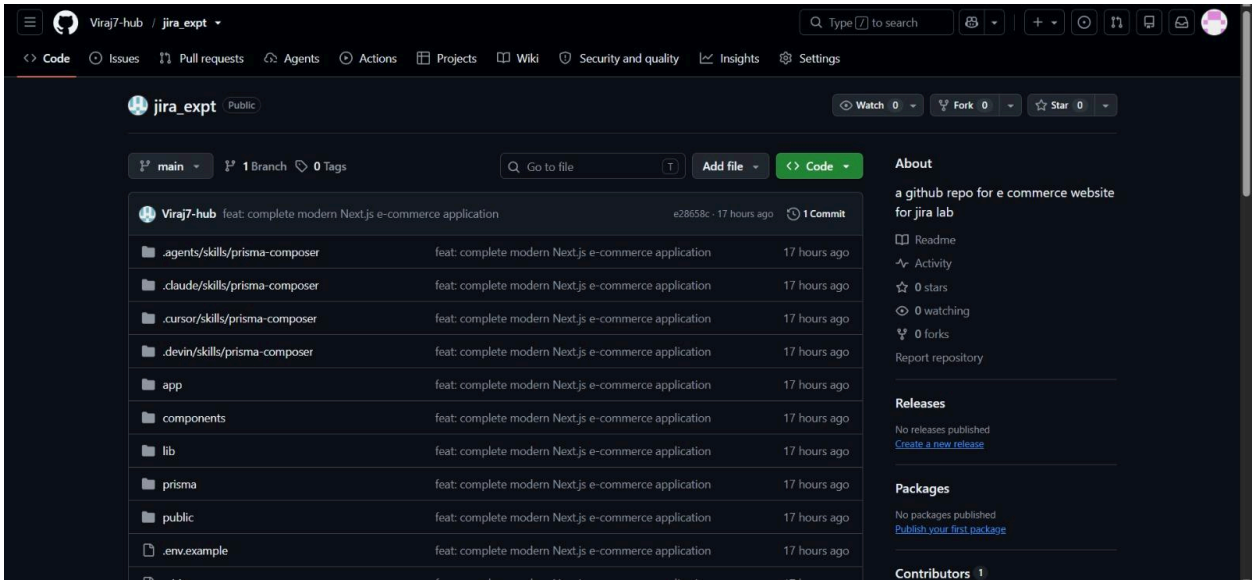
If you want, I can next add a small feature enhancement like product sorting, search, or an admin product creation form.

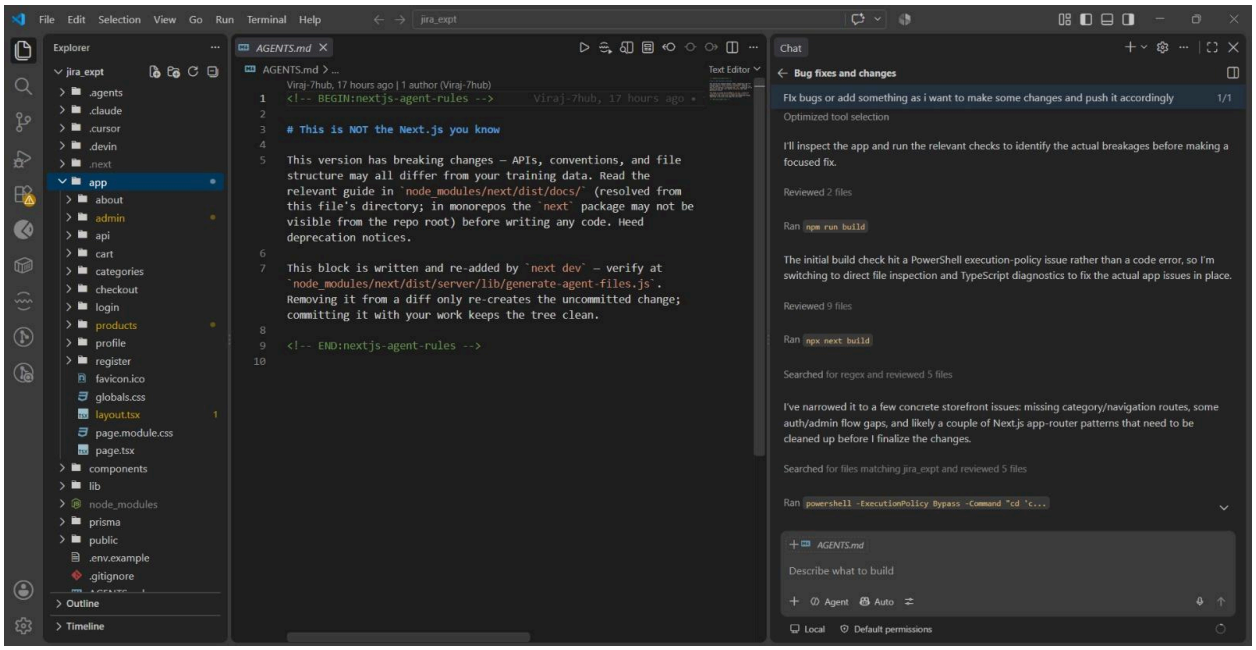
4:21 PM • MAI-Code-1.1-Flash • 10.0 credits




Chat
Bug fixes and changes
FIx bugs or add something as i want to make some changes and push it accordingly
1/1
Ran powershell -ExecutionPolicy Bypass -Command "cd 'c...
Reviewed 4 files
Ran git status --short --branch ; echo '---' ; git rem...
Reviewed 2 files
I've confirmed the root issue: several nav links point to routes that don't exist, and the product page ignores category query params. I'm fixing those missing routes and tightening the admin/login flow so the storefront behaves consistently.
Created 3 files and updated page.tsx +32 -14
I caught one more issue in the category filter: the original select lived in a server component, which breaks because browser-only actions like window.location and onChange won't work there. I'm converting it to URL-based navigation so it works cleanly in Next.js App Router.
Updated 2 files +63 -21
Ran git status --short --branch ; echo '---' ; git rem...
Ran powershell -ExecutionPolicy Bypass -Command "cd 'c...


Chat
Bug fixes and changes
FIx bugs or add something as i want to make some changes and push it accordingly
1/1
The repo is ready for a real push, and I'm validating it the correct way now by installing the project dependencies so the final checks reflect the actual app state rather than an empty workspace.
Ran powershell -ExecutionPolicy Bypass -Command "cd 'c...
Ran cd 'c:\Users\tanma\Downloads\jira_expt'; if (Test-...
Ran Set-ExecutionPolicy -Scope Process -ExecutionPolic...
Ran Set-ExecutionPolicy -Scope Process -ExecutionPolic...
Updated layout.tsx +2 -1
Ran Set-ExecutionPolicy -Scope Process -ExecutionPolic...
Ran Set-ExecutionPolicy -Scope Process -ExecutionPolic...
Ran cd 'c:\Users\tanma\Downloads\jira_expt'; git add a...
Optimized tool selection
Ran cd 'c:\Users\tanma\Downloads\jira_expt'; git push ...






```
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\tanma\Downloads\jira_expt>cd "c:\Users\tanma\Downloads\jira_expt"

c:\Users\tanma\Downloads\jira_expt>npm install

> e-commerce@0.1.0 postinstall
> prisma skills sync || exit 0
```

```
up to date, audited 372 packages in 7s

146 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

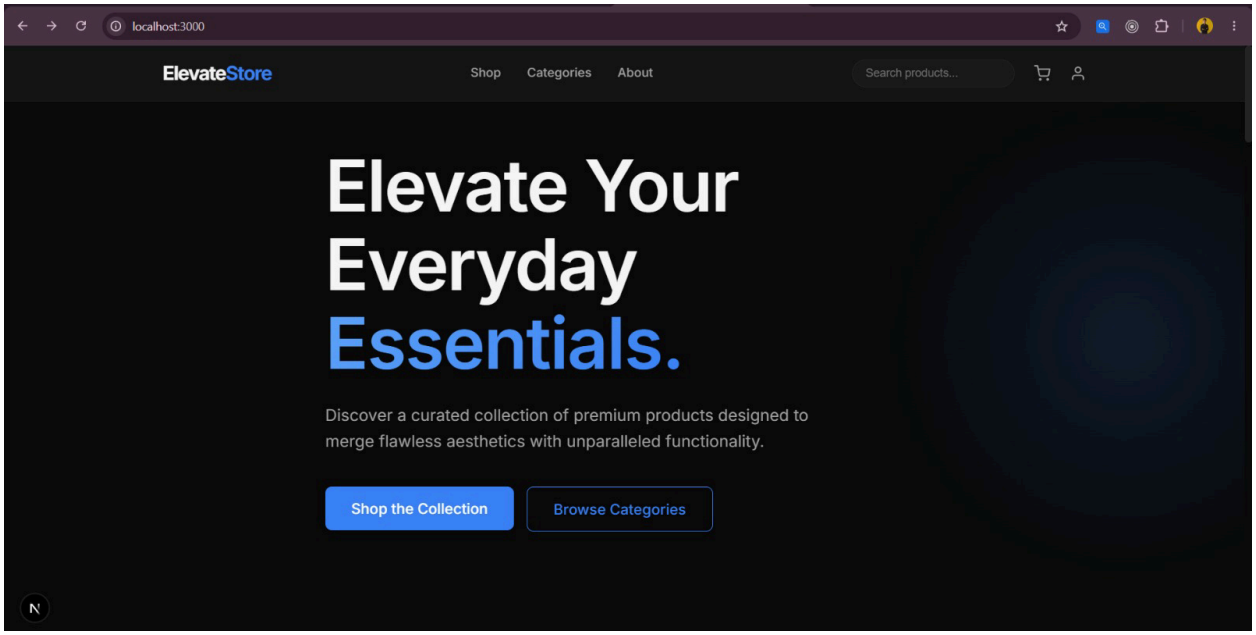
c:\Users\tanma\Downloads\jira_expt>npm run dev

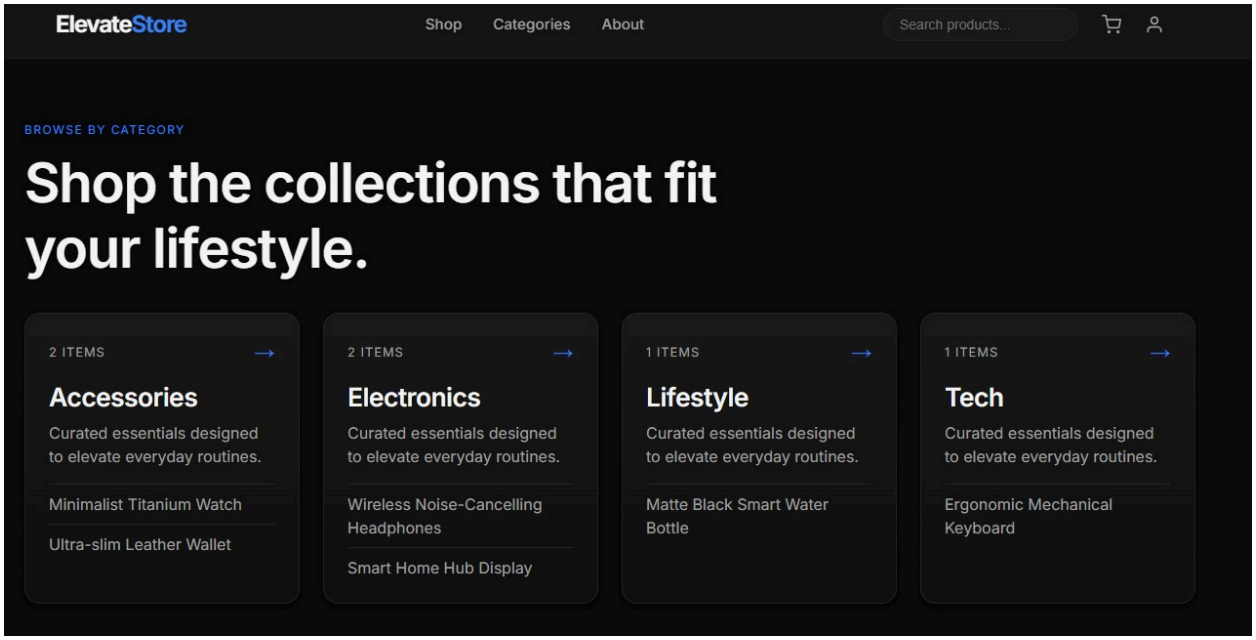
> e-commerce@0.1.0 dev
> next dev

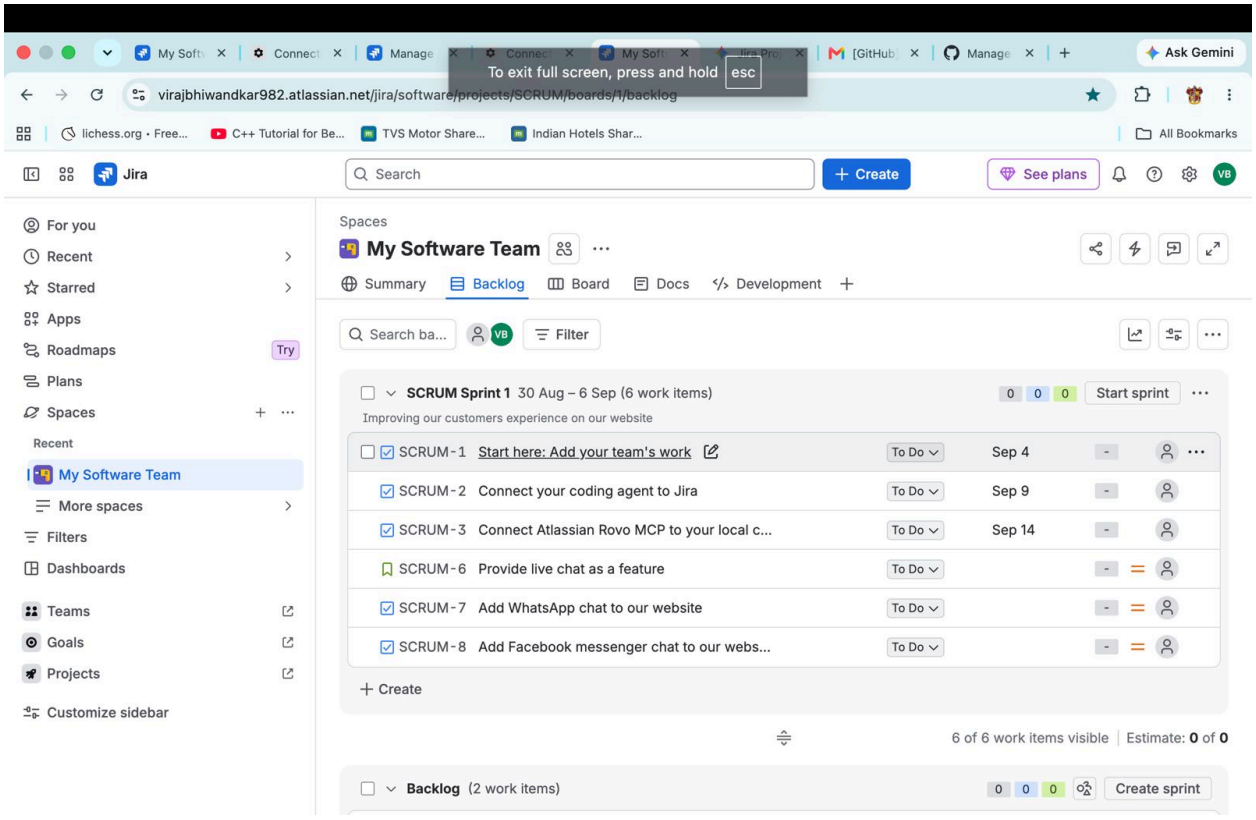
▲ Next.js 16.3.3 (Turbopack)
- Local:         http://localhost:3000
- Network:       http://192.168.56.1:3000
✓ Ready in 4.9s
✓ Running next.config.ts took 512ms

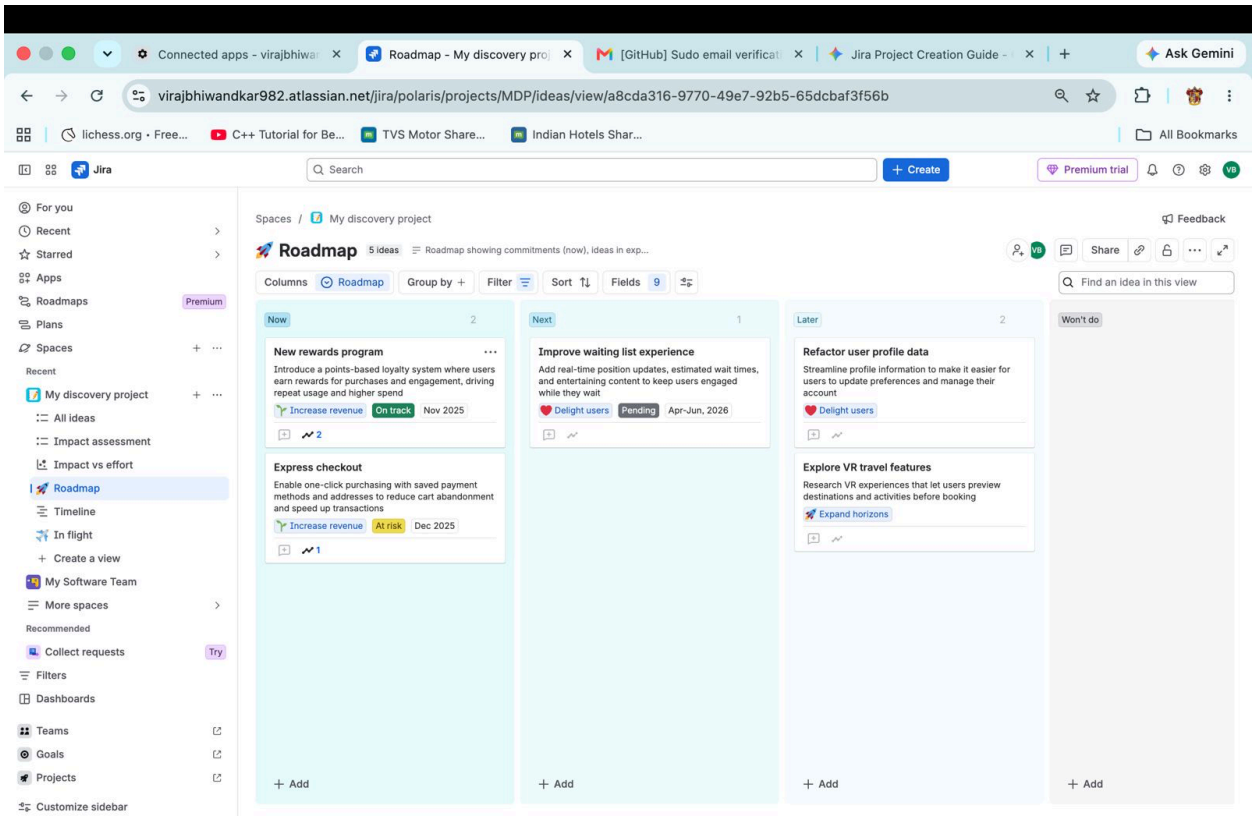
o Compiling / ...
 GET / 200 in 18.4s (next.js: 16.0s, application-code: 2.4s)
```

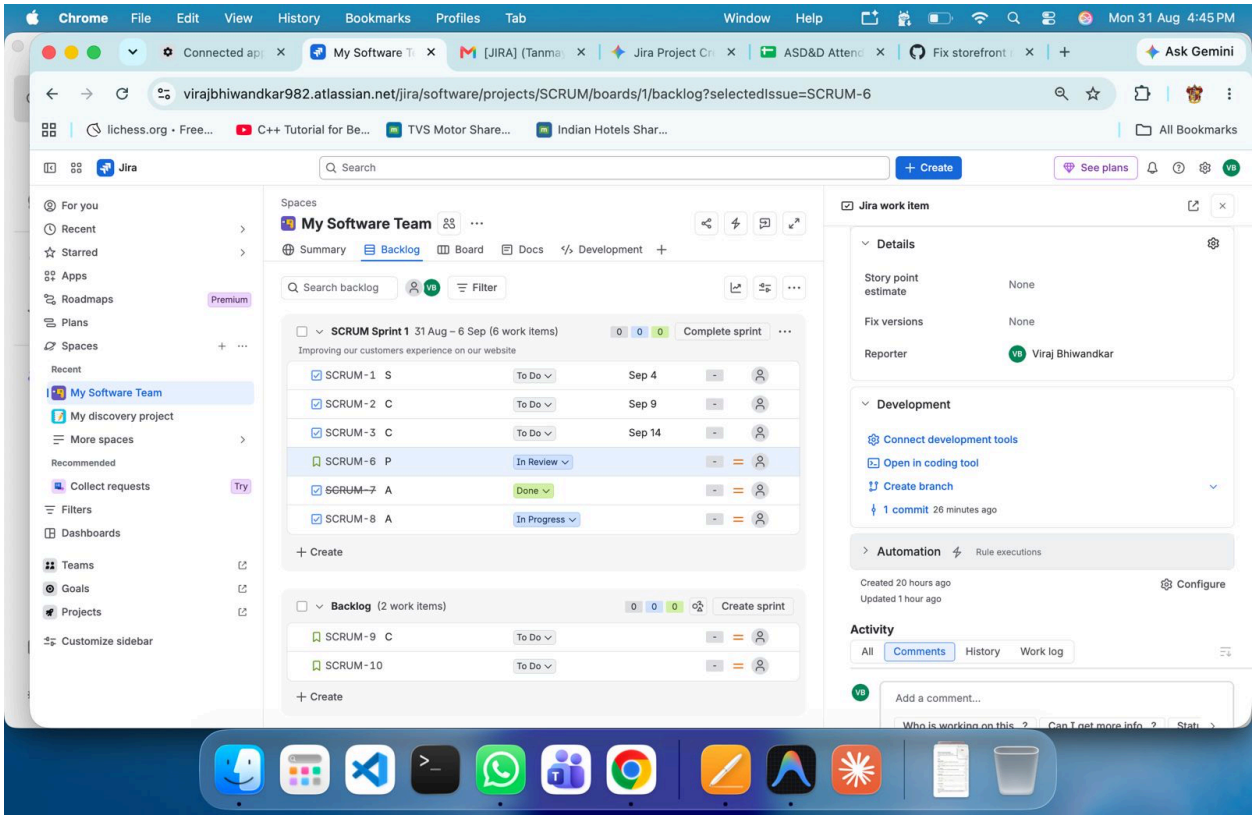
```
}
 GET /products 200 in 3.3s (next.js: 2.5s, application-code: 765ms)
⨯ upstream image response timed out for https://images.unsplash.com/photo-1595225476474-87563907a212?ixlib=rb-4.0.3&auto=format&fit=crop&w=800&q=80
✓ Compiled in 245ms
 GET / 200 in 311ms (next.js: 83ms, application-code: 228ms)
```

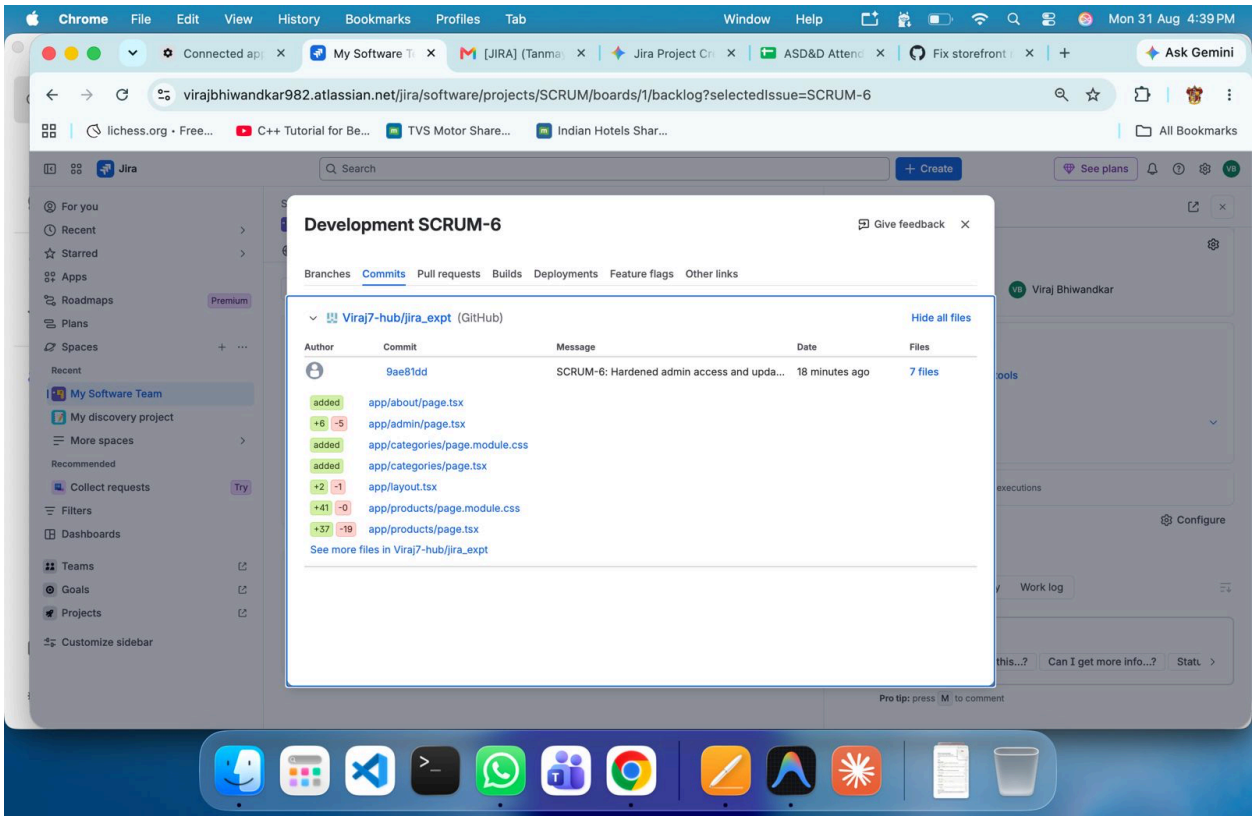


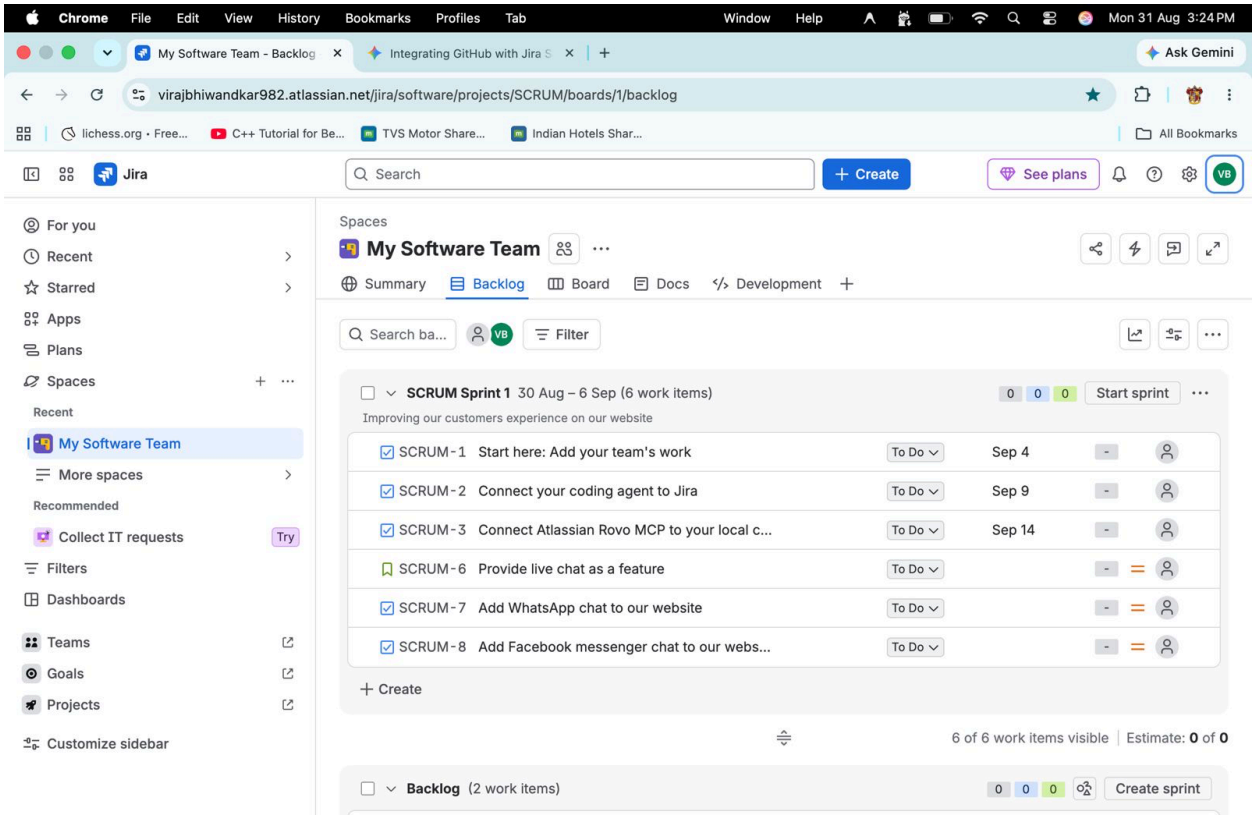

ElevateStore
Shop
Categories
About
Search products...
BROWSE BY CATEGORY
Shop the collections that fit your lifestyle.
2 ITEMS
Accessories
Curated essentials designed to elevate everyday routines.
Minimalist Titanium Watch
Ultra-slim Leather Wallet
2 ITEMS
Electronics
Curated essentials designed to elevate everyday routines.
Wireless Noise-Cancelling Headphones
Smart Home Hub Display
1 ITEMS
Lifestyle
Curated essentials designed to elevate everyday routines.
Matte Black Smart Water Bottle
1 ITEMS
Tech
Curated essentials designed to elevate everyday routines.
Ergonomic Mechanical Keyboard


To exit full screen, press and hold esc
virajbhiwandkar982.atlassian.net/jira/software/projects/SCRUM/boards/1/backlog
Ask Gemini
Jira
Search
Create
See plans
For you
Recent
Starred
Apps
Roadmaps
Plans
Spaces
My Software Team
More spaces
Filters
Dashboards
Teams
Goals
Projects
Customize sidebar
Spaces
My Software Team
Summary
Backlog
Board
Docs
Development
Filter
SCRUM Sprint 1 30 Aug – 6 Sep (6 work items)
Start sprint
Improving our customers experience on our website
SCRUM-1 Start here: Add your team's work To Do Sep 4
SCRUM-2 Connect your coding agent to Jira To Do Sep 9
SCRUM-3 Connect Atlassian Rovo MCP to your local c... To Do Sep 14
SCRUM-6 Provide live chat as a feature To Do
SCRUM-7 Add WhatsApp chat to our website To Do
SCRUM-8 Add Facebook messenger chat to our webs... To Do
Create
6 of 6 work items visible | Estimate: 0 of 0
Backlog (2 work items)
Create sprint


Roadmap - My discovery pro
virajbhiwandkar982.atlassian.net/jira/polaris/projects/MDP/ideas/view/a8cda316-9770-49e7-92b5-65dcbaf3f56b
Jira
Search
Create
Premium trial
Spaces / My discovery project
Feedback
Roadmap
5 ideas
Share
Columns
Roadmap
Group by
Filter
Sort
Fields 9
Find an idea in this view
Now
Next
Later
Won't do
New rewards program
Introduce a points-based loyalty system where users earn rewards for purchases and engagement, driving repeat usage and higher spend
Increase revenue
On track
Nov 2025
Improve waiting list experience
Add real-time position updates, estimated wait times, and entertaining content to keep users engaged while they wait
Delight users
Pending
Apr-Jun, 2026
Refactor user profile data
Streamline profile information to make it easier for users to update preferences and manage their account
Delight users
Express checkout
Enable one-click purchasing with saved payment methods and addresses to reduce cart abandonment and speed up transactions
Increase revenue
At risk
Dec 2025
Explore VR travel features
Research VR experiences that let users preview destinations and activities before booking
Expand horizons
Add
For you
Recent
Starred
Apps
Roadmaps
Premium
Plans
Spaces
My discovery project
All ideas
Impact assessment
Impact vs effort
Roadmap
Timeline
In flight
Create a view
My Software Team
More spaces
Recommended
Collect requests
Try
Filters
Dashboards
Teams
Goals
Projects
Customize sidebar


Chrome File Edit View History Bookmarks Profiles Tab Window Help Mon 31 Aug 4:45 PM
virajbhiwandkar982.atlassian.net/jira/software/projects/SCRUM/boards/1/backlog?selectedIssue=SCRUM-6
Jira
Search
Create
See plans
Spaces
My Software Team
Summary
Backlog
Board
Docs
Development
Search backlog
Filter
SCRUM Sprint 1 31 Aug – 6 Sep (6 work items)
Complete sprint
Improving our customers experience on our website
SCRUM-1 S To Do Sep 4
SCRUM-2 C To Do Sep 9
SCRUM-3 C To Do Sep 14
SCRUM-6 P In Review
SCRUM-7 A Done
SCRUM-8 A In Progress
Create
Backlog (2 work items)
Create sprint
SCRUM-9 C To Do
SCRUM-10 To Do
Create
Jira work item
Details
Story point estimate None
Fix versions None
Reporter Viraj Bhiwandkar
Development
Connect development tools
Open in coding tool
Create branch
1 commit 26 minutes ago
Automation Rule executions
Created 20 hours ago
Updated 1 hour ago
Configure
Activity
All Comments History Work log
Add a comment...
For you
Recent
Starred
Apps
Roadmaps
Premium
Plans
Spaces
My Software Team
My discovery project
More spaces
Recommended
Collect requests
Try
Filters
Dashboards
Teams
Goals
Projects
Customize sidebar


Development SCRUM-6
Give feedback
Branches Commits Pull requests Builds Deployments Feature flags Other links
Viraj7-hub/jira_expt (GitHub)
Hide all files
Author Commit Message Date Files
9ae81dd SCRUM-6: Hardened admin access and upda... 18 minutes ago 7 files
added app/about/page.tsx
+6 -5 app/admin/page.tsx
added app/categories/page.module.css
added app/categories/page.tsx
+2 -1 app/layout.tsx
+41 -0 app/products/page.module.css
+37 -19 app/products/page.tsx
See more files in Viraj7-hub/jira_expt


My Software Team
Summary Backlog Board Docs Development
SCRUM Sprint 1 30 Aug – 6 Sep (6 work items)
Start sprint
Improving our customers experience on our website
SCRUM-1 Start here: Add your team's work To Do Sep 4
SCRUM-2 Connect your coding agent to Jira To Do Sep 9
SCRUM-3 Connect Atlassian Rovo MCP to your local c... To Do Sep 14
SCRUM-6 Provide live chat as a feature To Do
SCRUM-7 Add WhatsApp chat to our website To Do
SCRUM-8 Add Facebook messenger chat to our webs... To Do
Create
6 of 6 work items visible | Estimate: 0 of 0
Backlog (2 work items)
Create sprint